



Design Checklist 01

Use this checklist to guide the design of starships. Although this checklist provides a sequence, steps may be completed in any order. The major number divisions in this checklist correspond to the governing Ship Design Charts.

The Ship FillForm (Charts 22 - 23)

Create a blank FillForm and use it to record design decisions. Enter the appropriate information (Code, Crew, CP Control Panels, Tons, Sq Squares, and Cost). If the entry is a TL other than the building TL, enter TL= N in Comments.

- A. Select a **name** for the ship.
- B. Identify planned tonnage.
- C. Identify the shipyard and TL.

01 Checklist. This checklist.

02 Select Mission for the ship.

- A. State the Service.
- B. State the intended Activity.
- C. State the Type of Activity.
- D. Indicate a Qualifier.
- E. Mission including Mission Code.
- F. Mission Modifiers.
- Q. Note (any) Small Craft Mission
- V. Note (any) Vehicle Mission.

03 Confirm Mission Definition.

04 Begin Hull Creation.

- A. Select Hull Size.
- B. Subhulls and Pods (as required)

05 Begin Hull Creation.

- C. Select Hull Configuration.
Friction, Agility, Stability.

06 Continue Hull Creation.

- D. Hull Structure. Base AV.
- E. Add Fittings.

07 Continue Hull Creation.

- F. Select Jump Field.
- G. Safe Jump Distance.
- H. Interference.

08-09 Armor

- A. Assign Armor Layers.
- B. Identify Armor Layer Type.
- C. Assign Anti-Layers.
- D. Assign (as required) Coating.

10-11 Starship Drives.

Note constructing Tech Level.
Select drives as required.

InSystem Interplanetary Drives:

M G R H

Interstellar Drives

J H S N

Power Systems

P U A C

X. Note Drive Efficiency.

12 Drive Potential.

Calculate the Drive Potential for each installed Drive.

13 Fuel Requirements.

Note InSystem Fuel Required

M G

Note Thrust Fuel Required

R H

Note Interstellar Drive Fuel Required

J H S N

Note Power System Fuel Required:

P U A C

F. Fittings

14-15 Sensors.

- A. Select one or more Sensors.
- B. Apply TL Stage Effects.
- C. Select a Mount.
- D. Mod Space Sensors for S=.
- E. Mod World Sensors for R=.
- F. Install a Sensor Control Panel.

16-17 Weapons.

- A. Select one or more Weapons.
- B. Apply TL Stage Effects.
- C. Select a Weapon Mount.
- D. Mod Space Weapons for S=.
- E. Mod World Weapons for R=.
- F. Install a Weapon Control Panel.

18-19 Defenses.

- A. Select one or more Defenses.
- B. Apply TL Stage Effects.
- C. Select a Defense Mount.
- D. Mod Space Defenses for S=.
- E. Mod World Defenses for R=.
- F. Install a Defense Control Panel.

20-21 Operations

- A. Ship's Troops and quarters.
- B. Specialists in Crew Structure
- C. Vehicles for the Ship.
- D. Small Craft for the Ship.
- E. Allocate Life Support.
- F. Assign Fuel Storage.
- G. Install Hull Connectors.

22 Controls

- A. Note available Control Panels.
- B. Install Consoles.
- C. Determine Staffing Level.
- D. Install Ship's Computer.

23 Payload.

- A. Available Payload: Tonnage.
- B. Assign Tonnage: Passengers.
 1. High and Middle.
 2. Low and Steerage.
- C. Crew Accommodations
- D. Life Support
- E. Access
- F. Cargo Hold capacity.

24-25 Crew Structure.

- A. Determine Total Crew Members.
- B. Crew Accommodation Tonnage.

26 Evaluate Ship Conditions.

- A. Crew and passenger Tonnage.
- B. Passenger Demand.
- C. Crew Comfort.
- D. Control Ergonomics.

27 Create the Ship QSP.

**Ship FillForm Part 1-2-3
Ship Combat Card**



02 Starship Missions

Starships are created to accomplish missions. The descriptive terms for missions have specific meanings, but they are defined to allow broad interpretation and substantial overlap.

Select a mission for the ship (or assign it after the ship is designed)

State the mission as a one-, two- or (rarely) three- letter code.

Multiple identical letter codes (AA, AAA) may use a digit (A2, A3)

Apply Mission Modifiers after the Mission as needed in the format EFF. Although the technically correct usage states Mission first, the converse has become commonplace as FFE. and is tolerated: Modifier-Modifier-Mission (SDB= System Defense-Boat). Actual meanings are subject to common sense, and may be ambiguous.

Select An ACS Ship Mission

A Service	B Activity	C Type	D Qualifier	E Mission
Naval	Combat	Offensive	Principal	Cruiser C
			Major	Frigate G
			Special	Destroyer V
			Minor	Corvette E
		Siege	Attack	Ortillery H
			Invasion	Assault T
			Defender	Sentinel S
	Defensive		Minor	Escort E
			Special	Boat B
			Major	Defender D
		Independent	Principal	Monitor N
			Anti-Shipping	Corsair P
			Anti-Commerce	Raider R
	Auxiliary Supply		Anti-Port	Marauder P
			Major	Transport T
			Minor	Barge W
			Resupply	Tender, Tug T
			Information	Corvette E
Commerce	Merchant	Scheduled	Passenger	Liner M
			Cargo	Merchant R
			Freight	Freighter F
		UnScheduled	Freight	Transport T
			Cargo	Trader A
			Passenger	Packet U
	Charter		Recreation	Safari K
			Active	Expedition K
			Luxury	Yacht Y
		Information	Small Goods	Courier S
			Data Files	Messenger S
			Goods and Files	Express X
Government NGO Private	Exploration		First Look	Scout S
			Re-Look	Survey N
			Data Collection	Beagle B
			Medical Data	Med N
			Data Analysis	Lab L
			Resource Search	Prospector J
	Bureaucratic		Inspection	Picket P
			Enforcement	Patrol P
				Privateer P
Unclassified				Z

The following terms apply consistently to ship missions.

Freight. Goods carried for a fee; Ownership remains with the shipper.

Cargo. Goods owned by the ship; carried for ultimate sale at a profit.

Passengers. Individuals who travel on a ship in exchange for a fare.

Select Mission Modifiers

F Modifier	
Alternate, Improved, Armored, Attack.	A
Boat. Bulk. Battle. Big.	B
Close. Carrier. Communications.	C
Defense. Defending. Interceptor.	D
Escort. Essential. Electronic Warfare.	E
Fast. Fat. Frontier. Far. Flag. Free.	F
Gunned. Upgunned. Gas.	G
Fuel. Tanker. Hydrogen.	H
Survey. Prospector. Interface. Intruder.	J
Subsidized. Fast, Diplomatic.	K
LR. Lifeboat. Exploratory. Light.	L
Military. Militia. Mercenary. Motivator. Tug.	M
Naval. Nuclear. Fleet.	N
Patrol. Plus. Passenger, Mercenary.	P
Disguised. Decoy. Quarantine, Mother.	Q
Recon. Rescue. Rider.	R
Slow. System. Special. Luxury. Small.	S
Tramp. Tender. Transport.	T
Unarmed, Hulk. De-activated. Inop.	U
Vehicle Carrier, Drone. Remote.	V
Unpowered. Non-Jump.	W
Experimental. Special. Express.	X
Hull. Subhull. Pod. Rider. Modular.	Y
Unassigned.	Z

Q SMALL CRAFT

Lifepod	10	A
Fighter	10	F
Gig	20	G
Launch	20	L
Ship's Boat	30	B
Pinnace	40	N
Cutter	50	C
Cutter Module	30	M
Shuttle	95	S
Pod	< 100	P
Drone	varies	D

V VEHICLE

Air/Raft	R
Flyer	F
Ground	G
-ATV	A
-Car	C
-Truck	T
-Vehicle	V
-Mover	M
WaterCraft	W
-Boat	B
-Sub	U

SHIP MISSION FORMAT FFE (examples)

BC =Battle Cruiser

AG =Attack Frigate

LGR =Long-Range Upgunned Raider



Starship Missions 03

Fighting Ships

Fighting ships are ranked **Principal** or **Line** (the best available), **Major** (better than most) and **Minor** (merely acceptable).

A **Cruiser** is the principal type of offensive fighting ship (in both weapons and defenses). A **Frigate** (although less than a cruiser) is a major type. A **Corvette** is a minor type (a small craft in this mission is an Attack **Fighter**).

A **Monitor** is the principal type of defensive fighting ship (favoring armor over firepower). A **Defender** is a major defensive ship; an **Escort** is a minor type (a small craft in this mission is an Interceptor **Fighter**).

Ships for operations against worlds include **Ortillery** for firepower and bombardment, **Assault** to deliver troops and equipment, and **Sentinel** to defend the system from invaders.

Independent naval forces include: **Corsair** to attack commercial shipping (specifically to steal its freight or cargo). **Raider** to attack and destroy commercial shipping. **Marauder** to attack ports and installations.

Auxiliaries

Naval auxiliaries are not intended for combat. A **Transport** is a major multi-purpose ship carrying supplies and personnel (a small craft in this mission is a **Shuttle**). A **Barge** is an unpowered support hull (a small craft in this mission is a **Pod**). A **Tender** provides supplies and support to naval vessels.

A **Corvette** is often pressed into auxiliary service as a courier.

Merchant Ships

Merchants are private business vessels. On scheduled routes: a **Liner** carries primarily passengers, a **Freighter** carries primarily freight, a **Merchant** carries primarily cargo.

For unscheduled routes: a **Packet** carries primarily passengers, a **Trader** carries primarily cargo, a **Transport** carries primarily freight.

An **Expedition** is chartered for travel to unexplored locations. A **Safari** is chartered to visit world surfaces for recreation and tourism.

A **Yacht** is a privately owned and operated pleasure ship.

Couriers

A **Courier** is built to carry small shipments of important goods. A **Messenger** carries communications or data files. An **Express** carries both goods and files (and the name implies extreme speed). A **Corvette** is a naval courier.

Explorers

A **Scout** is a small first-in ship compiling star system basic information. A **Survey** periodically visits known systems to gather or confirm data. A **Beagle** visits new systems to gather detailed scientific data.

A **Med** or **Med Ship** gathers data affecting sophont health. A **Lab** or **Laboratory Ship** gathers scientific data with the ability to process it immediately.

A **Prospector** conducts planetological (or planetoidological) exploration, usually searching for exploitable resources.

Bureaucratic

A **Picket** is a sentry charged with detection and inspection of ship traffic.

A **Patrol** enforces traffic and commerce regulations.

Small Craft

Small Craft are less than 100 tons. Traditionally accepted small craft types (and suggested tonnages) are:

Pod (< 100 tons). Unpowered hull.

Lifepod (10 tons). Emergency escape and survival craft.

Fighter (10 tons). Armed (and possibly armored) combat craft.

Launch (20 tons). Small multi-purpose craft capable of landing on worlds.

Gig (20 tons). Utility craft for passenger and freight transfer between ships or bases.

Ship's Boat (30 tons). Mid-sized utility craft assigned to a specific ship.

Pinnace (40 tons). Mid-sized utility craft capable of landing on worlds.

Cutter (50 tons). Mid-sized multi-purpose small craft.

Shuttle (90 tons). Large passenger and freight transporter capable of landing on worlds. Built to travel between two points (worlds; locations).

Drone (varies). Remotely operated or controlled.

Mission Modifiers

Modifiers are applied as needed.

Alternate. Differs from normal.

Armored. Up-Armored. Armor is a substantial addition.

Battle. Includes added firepower.

Boat. No interstellar drives.

Bulk. Payload is solid (granular), liquid, or gas bulk freight.

Carrier. Equipped to enclose and transport ships, vehicles, or forces.

Close. Operates closely with other ships (typically the same Space Range).

Defense. Defending. Equipped for anti-intruder missions.

Disguised. Misleads adversaries.

Escort. For defense missions.

Experimental. Incorporates untried or untested technology.

Far. Greater than typical Jump.

Fast. Greater G Capability.

Fat. Higher hold capacity.

Frontier. For remote areas.

Fuel. Tanker. Fuel-related.

Gunned. More firepower.

Hydrogen. Fuel related.

Improved. Added performance.

Intruder. Probes enemy territory.

Light. Smaller; less tonnage.

Local. System. Operates within a single system.

Long Range. LR. Sustained time or distance operations.

Mercenary. Military for hire.

Military. Militia. Supports Army.

Minus. Inferior capabilities.

Modular. Uses modules.

Naval. Supports Naval operations.

Plus. Specialized. Some capability has been enhanced.

Passenger. Carries people for hire.

Patrol. Operates within a defined region, but without scheduled routes.

Recon. Capable of military or naval intelligence gathering.

Slow. Lower G capability.

Standoff. Operates at Long Range.

Subhull. Component hull.

Subsidized. Operating costs are primarily paid by government.

Tender. Resupplies other ships.

Tramp. Flies unscheduled routes.

Unarmed. No weapons installed.



04 Hull Tons and Costs

Hull Size (in tons) has a cost determined by tonnage and configuration. It is possible to create a ship with more than one hull.

CREATING A HULL (AB)

- Select a **Hull Size** to determine volume (and payload).
- Select **Component Hulls** (Pods, Barges, or Subhulls).

A HULL COSTS

-----Configuration Costs MCr-----									Total	Total
ID	Tons	C	B	P	U	S	A	L	Squares	Cubes
A	100	2	3	1	5	8	9	16	200	400
B	200	4	6	2	8	14	16	28	400	800
C	300	6	9	3	11	20	23	40	600	1200
D	400	8	12	4	14	26	30	52	800	1600
E	500	10	15	5	17	32	37	64	1000	2000
F	600	12	18	6	20	38	44	76	1200	2400
G	700	14	21	7	23	44	51	88	1400	2800
H	800	16	24	8	26	50	58	100	1600	3200
J	900	18	27	9	29	56	65	112	1800	3600
K	1000	20	30	10	32	62	72	124	2000	4000
L	1100	22	33	11	35	68	79	136	2200	4400
M	1200	24	36	12	38	74	86	148	2400	4800
N	1300	26	39	13	41	80	93	160	2600	5200
P	1400	28	42	14	44	86	100	172	2800	5600
Q	1500	30	45	15	47	92	107	184	3000	6000
R	1600	32	48	16	50	98	114	196	3200	6400
S	1700	34	51	17	53	104	121	208	3400	6800
T	1800	36	54	18	56	110	128	220	3600	7200
U	1900	38	57	19	59	116	135	232	3800	7600
V	2000	40	60	20	62	122	142	244	4000	8000
W	2100	42	63	21	65	128	149	256	4200	8400
X	2200	44	66	22	68	134	156	268	4400	8800
Y	2300	46	69	23	71	140	163	280	4600	9200
Z	2400	48	72	24	74	146	170	292	4800	9600

ID= Hull Identifier (or Hull Code). Cost in MCr for the final hull.

Barges and Subhulls are also created using this table

B POD COSTS

-----Configuration Costs MCr-----									Total	Total
ID	Tons	C	B	P	U	S	A	L	Sq	Cubes
A1	10	.2	.3	.1	.5	.8	*	1.6	20	40
A2	20	.4	.6	.2	.8	1.4	*	2.8	40	80
A3	30	.6	.9	.3	1.1	2.0	*	4.0	60	120
A4	40	.8	1.2	.4	1.4	2.6	*	5.2	80	160
A5	50	1.1	1.6	.5	1.7	3.2	*	6.4	100	200
A6	60	1.3	1.9	.6	2.0	3.8	*	7.6	120	240
A7	70	1.5	2.2	.7	2.3	4.4	*	8.8	140	280
A8	80	1.7	2.5	.8	2.6	5.0	*	10.0	160	320
A9	90	2.0	2.9	.9	2.9	5.6	*	11.2	180	360

ID= Hull Identifier (or Hull Code).

* If Parent Hull is Airframe, use Streamlined S on this chart.

Configuration. If a Pod or Barge has a configuration different from the parent hull, the least beneficial set of capabilities is used.

Jump Fields. If the parent hull has Jump Grid, then the Hull Pod must have Jump Grid.

Starship Tons

Hulls are built in 100 ton increments ranging from 100 to 2400 tons.

Hull sizes are expressed in displacement or volume Tons (= 13.5 cubic meters).

Ultimate Hull Tonnage May Vary. Undertonnage improves performance. Overtonnage reduces performance. Gross overtonnage (more than 49 tons) is rounded to the next larger hull.

Hardpoints

Hardpoints allow the installation of weapons and other important support items.

A Hull has one Hardpoint per 100 tons: a limit on the number of weapons that can be installed.

PODS AND BARGES

Pods and Barges are detachable ship components. Adding a Pod or Barge to a parent Hull increases the total tonnage for a ship, which potentially decreases the performance of its drives. Detaching a Pod or Barge from a Parent Hull decreases the total tonnage for a ship, potentially increasing its performance.

Building Barge And Pod Hulls

Hulls may have detachable components.

A **Pod** is a detachable unit < 100 tons.

A **Barge** is a detachable hull (100+ tons) usually unpowered and incapable of independent action.

A **Subhull** is a component hull (100+ tons) which may be detachable.

Pods are created from the Pods Table; Barges and Subhulls are created on the Hulls table.

Pods, Barges, and Subhulls are attached to the parent hull with interlocking clamps called **Grapples**. One Grapple set consists of two 1-ton Grapples (one on the Pod, Barge, or Subhull; one on the parent hull). One set is required for each 35 tons of Pod, Barge, or Subhull.

DECK PLANS FOR STARSHIPS

It is possible to create deck plans for starships based on their tonnage.

Deck Squares. One ton equals two deck squares when creating deck plans (assuming a 3 meter deck separation).

Deck Cubes. One cube is 0.25 tons. One ton equals four 1.5 meter cubes.



Starship Configurations 05

Configuration determines ship appearance and performance.

CREATING A HULL (C)

C. Select a Configuration and note ship performance characteristics including Friction, Agility, and Stability.

C HULL CONFIGURATION CHARACTERISTICS

Type	Friction	Agility	Accel	Max G	Stability	Land	Comment
C Cluster	x2	-5	--	1	-3	No	An accumulation of compartments.
B Braced	x2	-4	--	3	-2	No	A cluster structured to allow higher acceleration.
P Planetoid	x1	-2	--	9	-1	?	A hollowed nickel-iron asteroid.
U Unstreamlined	/2	-1	--	9	0	Yes	An enclosure whose protrusions increase drag.
S Streamlined	/3	0	--	9	+1	Yes	An enclosure with cowlings and fairings to decrease drag.
A Airframe	/4	+1	+1	9	+2	Yes	A winged enclosure for better performance in atmosphere.
L Lifting Body	/5	+1	+1	9	+3	Yes	A radically streamlined lifting-surface body

Hull Characteristics

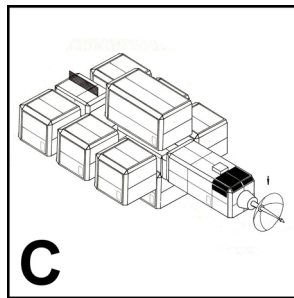
Agility. Mod on maneuver in atmosphere.

Stability. Mod on Turbulence when operating in Atmosphere.

Friction. Multiplier on Boost and Reentry friction heating.

Acceleration. Wing-based Mod (an increase) on Acceleration by maneuver drives in atmosphere. A 1G drive on a winged hull can produce 2G.

Max G. Maximum G a hull can withstand based on structure and the limits of inertial compensators.

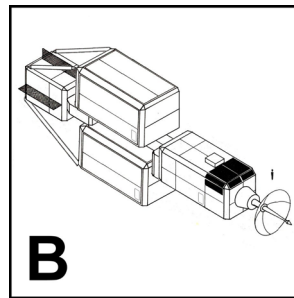


C

Cluster Hull

An accumulation of compartments and various connectors.

Least expensive of the hull types. Unsuitable for high acceleration; not intended for world landings.

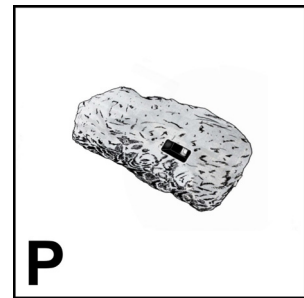


B

Braced Cluster Hull

An upgraded Cluster Hull braced to allow relatively higher accelerations.

Cluster Hull upgraded to allow higher acceleration. Not intended for world landings.



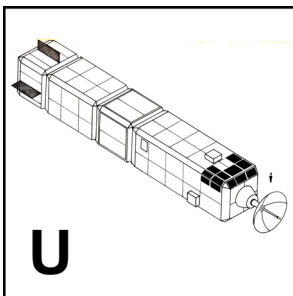
P

Planetoid Hull

A small planetoid with a laser-tunnelled interior for drives, crew, and payload.

Relatively inexpensive and structurally very strong.

Unsuitable for most world landings.



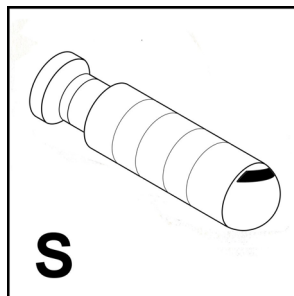
U

Unstreamlined Hull

A relatively compact hull marginally capable of entering atmosphere.

A commonly encountered hull type handicapped by its inattention to streamlining, but also cheaper.

Unsuitable for most world landings.



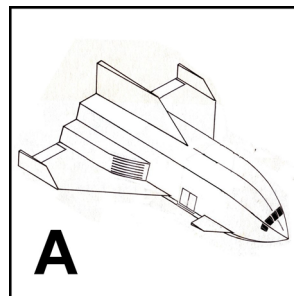
S

Streamlined Hull

A relatively compact hull crafted to support maneuver in atmospheres.

Common hull type designed for efficient atmospheric operations in addition to space travel.

Intended for landings on most worlds.



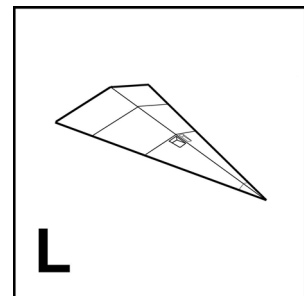
A

(Winged) Airframe Hull

A streamlined hull with wings and fins to enhance maneuver in atmosphere.

Streamlined hull with the wings and fins intended for efficient atmospheric operations.

Designed for operation in atmosphere.



L

Lifting Body Hull

A hull with integral lifting surfaces to support maneuver in atmosphere. Most efficient of the atmosphere-capable hulls.

Lifting Body receives the effect of Wings but not Fins.

Designed for operation in atmosphere.



06 Starship Hull Fittings

Hull Structure determines the ship's construction technique and its base armor. Fittings enhance ship performance.

CREATING A HULL (DE)

D. Select a Hull Structure to define the basic nature and construction technique of the Hull.

E. Add Fittings to allow for landing gear, flotation and submergence capability, and wings.

D HULL STRUCTURE

Type	Configuration	TL=	AV=	Comment	Features
A Plate	C B - U S A -	8	8 =TL	Standard/ default.	Also called Frame and Plate
S Shell	- - - S A L	9	5 =TL /2	Frameless alloy skin.	Lighter (Half Tonnage)
P Polymer	C B - U S A -	10	5 =TL /2	Specialized plastics.	Self-Healing
F FeN	- - P U - -	11	20	Hollowed nickel-iron asteroid.	Upgradable to U
O Organic	C B U S A -	12	6 =TL /2	Captured or farmed organisms.	Regenerating
C Charged	C B U S	14	28* =TLx2	Powered improvement of Plate	*AV= TLx2 when powered

Hull Structure is a naval architecture decision (the default is Plate).

Structure does not usually affect Hull tonnage, but Shell Hull tonnage is halved.

Structure does not usually affect Hull cost; but Organic Hull cost is halved; Charged Hull cost adds MCr1 per 100 tons.

E STARSHIP HULL FITTING OPTIONS

Fitting	C	B	P	U	S	A	L	Comment	TL	Tons**	MCr***
D Flotation Hull	No	No	+	+	+	+	=	Water Landing*	5	1.0	1.0
E Submergence Hull	No	No	+	+	+	+	+	Water Landing*	6	2.0	2.0
F Fins	No	No	No	+	+	=	+	Increases Agility in Atmosphere.	5	2.0	0.5
G Folding Fins	No	No	No	+	+	No	No	Increases Agility in Atmosphere.	8	0.0	0.5
W Wings	No	No	No	+	+	=	=	Increases performance in Atmos.	7	5.0	1.0
V Folding Wings	No	No	No	+	+	No	No	Increases performance in Atmos.	9	1.0	2.0
Landing Skids (Standard)	+	+	+	+	=	+	=	Tarmac Landing	7		
K Landing Legs with Pads	+	+	+	+	+	+	+	Allows Wilderness Landing	8	1.0	1.0
M Landing Wheels	+	+	+	+	+	=	+	Allows Glide Landing	5	3.0	1.5
Z Lifters	+	+	+	+	+	+	+	Allows limited hover and move	8	--	(0.5)

TL= Minimum TL to install fitting. Lifters are automatically installed (if TL is met). Reduce Hull Cost if they are not installed.

*Includes Glide Landing on Water. **Tons per 100 tons of hull. ***MCr per 100 tons of hull.

+ Installation allowed on this Configuration. = indicates that the hull already has this benefit. No: Installation not allowed.

Lifting Bodies have the benefit of Wings but not Fins (although Fins can be added).

Landing Gear

Landing Gear provides options for ship launch and landing capabilities.

Landing Skids. Retractable horizontal bars bear the ship's weight. Requires solid bedrock or tarmac landing site. Standard or default option.

K. Landing Legs With Pads. Retractable legs end in pads. Can tolerate uneven landing terrain.

M. Landing Wheels. Retractable legs end in wheels. Allows glide landing / takeoff from airstrip. Requires tarmac or bedrock landing site. Required if Wings are used for Liftoff and Landing.

Wings and Fins

Wings and Fins provide enhanced performance in World Atm 2+.

F. Fins. Fins increase Agility +1.

G. Folding Fins. Functions as Fins when deployed; can be folded. Folded Fins occupy no Tonnage.

V. Collapsing Wings. Function as Wings when deployed; can be collapsed. Collapsed Wings occupy no Tonnage.

W. Wings (includes Fins). Wings increase Maneuver or Gravitic Drive +1G.

Z. Lifters. Grav Plates provide limited hover capability and slow local movement (not NOP or fast maneuver).

Flotation

Flotation Status allows a hull or pod to float (whether in water or other environmental liquid).

D. Flotation Hull. Sealed against prolonged water or fluid exposure. Allows glide landing or takeoff from water.

E. Submergence Hull. Sealed against prolonged water or fluid exposure. Includes ability to submerge and resurface. Allows glide landing / takeoff from water. Doubles the Pressure the hull (based on its Armor) can withstand.



Starship Jump Fields 07

CREATING A HULL (FGH)

Jump Drives require a Jump Readiness option.

F. Jump Field Type. Choose a Jump Field option.

G. Safe Jump Distance. Calculate the distance.

H. Interference. Note formula to determine if Initiation Interference occurs.

JUMP BUBBLE

A **Jump Bubble** creates a spherical field around the ship and centered on the jump drive. Jump Bubble is the standard for generating a Jump Field; it does not interfere with armor and produces a standard jump flash.

Jump Bubble allows a ship to vary its effective tonnage from mission to mission (which makes Drop Tanks and Variable Jump Container Ships possible).

Difficulties. A Jump Bubble may enclose nearby debris, which may result in a Jump Mishap.

JUMP GRID

A closely-conforming **Jump Grid** channels jump energy through a mesh of conduits and cables embedded in the hull.

Jump Grid allows a reduced safe jump distance, making it possible for a ship to jump closer to a gravity source. On the other hand, the Jump Grid reduces the strength of armor and increases telltale jump flash.

Difficulties. Damaged Jump Grids are difficult to repair. (Jump Plates are bolted on to repair gaps in grid coverage, which converts it to Jump Plates).

JUMP PLATES (ALSO REPAIR PLATES)

The original system for creating a Jump Field consisted of Bolt-On Jump Plates (one plate per 10 tons of Hull; bolted on the surface in an evenly spaced pattern. Jump Plates are required for any Experimental, Prototype, or Early Tech Jump Drive (or Experimental, Prototype, or Early Hop, Skip, or Higher Order drive).

Jump Plates reduce armor value for the outer armor layer. They produce higher values of D.

Repair Plates. Jump Plates are used to repair damaged Jump Grids: they are bolted on damaged hull to restore the Jump Grid continuity. When in use, Repair Plates effectively convert Jump Grid to Jump Plates for Jump Field production.

The Hull Armor at the repaired Compartment of Hit Location is reduced by half.

Advantages. Jump Plates are a lower tech component of the Jump Drive. Jump Plates are a repair option and can be bolted over damage hull Jump Grids (reducing the overall to Plates).

F JUMP FIELD TYPES

Type	Strength	Comment	Armor Mod	Flash	
Bubble	100	Default	std	std	Bubble produces the standard value for D (Table 07G).
Grid	80	Embedded in the Hull	Mod -1D	+1	Grid reduces safe jump distance D.
Plates	140	1 Plate Per 10 Hull Tons	Mod /2	+1	Plates produce the greatest value for D.

Flash: Change in Jump Flash Size from Standard.

G CALCULATE SAFE JUMP DISTANCE

The safe jump distance D for a specific ship is

$$D = S / E - K$$

D= Diameters (from the Gravity Source Center).

S= Jump Field Strength (Chart 07F).

E= Drive Efficiency (from Chart 10X).

=1 unless Drive Tech Stage Effects are in place.

K= Engineer Skill and Jump Drive Knowledge.

For example, Fat Trader *March Harrier* with Jump Plates (S= 140), a Standard Jump Drive (E= 1) and Engineer Jump Knowledge-3. $D = 140 / 1 - 3 = 140 - 3 = 137$. This ship can safely Jump at 137 Diameters.

For example, Gunned Escort *Neon* with Jump Grid (S= 80), Advanced Jump Drive (E=120%= 1.2) and Engineer Jump Knowledge-5. $D = 80 / 1.2 - 5 = 66 - 5 = 61$. The ship can safely jump at 61 Diameters.

H INITIATION INTERFERENCE

The safe distance to avoid initiation interference is

$$X = S / E - (D + K) (+Flux)$$

X= Initiation Interference. If $X < 1$, no jump interference.

Flux is optional, but if rolled, must be used.

D= Diameters (from the Gravity Source Center).

S= Jump Field Strength (Chart 07F).

E= Drive Efficiency (from Chart 10X).

=1 unless Drive Tech Stage Effects are in place.

K= Engineer Skill and Jump Drive Knowledge.

For example, Fat Trader *March Harrier* (S=140, E=1, K=3), fleeing a security inspection, reaches 100D. $X = 140 / 1 - (100 + 3) = 140 - 103 = +37$. X is greater than zero: there will be a misjump if the ship attempts Jump (even Flux -5 can't change X to 0 or less).

For example, Gunned Escort *Neon* (S=80, E=1.2, K=5) reaches 70D as it approaches an outpost when it suddenly encounters an overwhelming enemy force. $X = (80 / 1.2) - (70 + 5) = 66 - 55 = -11$. X is less than 1; the ship can safely jump.



08 Starship Armor1

Base TL
equals
Building
Shipyard TL



TL
Armor
Blank



Armor
Generic

Armor		8		9		10		12		8		14	
	A		S		P		O		F		C		Protects Against
	Armor Plate		Armor Shell		Armor Polymer		Armor Organic		Armor FeN		Armor Charged		
AV	= TL		= TL / 2		= TL / 2		= TL / 2		= 20		= TL x 2		AV=Armor Value
BF	= TL x 10		= TL x 5		= TL x 5		= TL x 5		= 200		= TL x20		BF=Bang Flash
Pen	= TL		= TL		= TL		= TL		= 200		= TL x 2		Pen=Penetration
EMP	= TL x 10		= TL x 5		= TL x 5		= TL x 5		= 200		= TL x20		EMP=EM Pulse
Rad	= TL x 10		= TL x 5		= TL x20		= TL x 5		= 200		= TL x20		Rad=Radiation
Heat*	= TL x100		= TL x100		= TL / 2		= TL x 5		= 2000		= TL x20		Heat=Heat
Pres	= TL x 10		= TL x 10		= TL x20		= TL x 5		= 200		= TL x20		Pres=Pressure
*also			Half-Tonnage		Self-Healing		Regenerating		Planetoid				=Special Notes
Burn	CBUS		SAL		CBUSA		CBUSA		PU		CBUS		=Config

TL 6	Aluminum												
7		Titanium											
8	PLATE		Laminate		Chitin		FeNi						
9		SHELL											
10	Dense		POLYMER		Keratin								
11		CrystallIron											
12	Kinetic		Composite		ORGANIC						+Aluminum		
13		LiteMetal											
14	SuperDense		Bonded								CHARGED PLATE		
15		VLiteMetal											
16	HullMetal										+Dense		
17		Matrix					Geneered						
18											+Kinetic		
19													
20											+SuperDense		
21													
22											+HullMetal		

Alternative TL Armor Identifier Options

Names are assigned to the types of Armor, and those names themselves have TL and TL Stages.

For example, TL6 Plate Armor is called Aluminum (AV= 6). It can be further refined as Experimental Aluminum (TL3 AV=3), or Advanced Aluminum (TL 9 AV=9). For some cases, the protection values of a specific Armor Type and Name may duplicate the values of a different Type and Name. Tech Stage differences in cost and mass are ignored.

Coatings and Anti-Layers							
	N	P	Q	B	E	K	R
	Ablat Coat	Reflec Coat	Slick Coat	Anti-Blast	Anti-EMP	Anti-Kinetic	Anti-Rad
AV	= 0	= 0	= 0	= TL	= TL	= TL	= TL
	Heat= TL x100	Burn = TL x100	Trac = TL x100	BF = TL x10	EMP = TL x10	Pen= TLx10	Rad= TLx100



Starship Armor2 09

ARMOR LAYERS

Armor is determined by Hull Structure, applied in Layers, and may be customized against specific threats.

Armor Type. A Hull with Armor Type= Plate is constructed of Plate, which is the name of its Armor.

Cost. Armor imposes no additional monetary cost; its cost is the loss of available tonnage.

Layers. Armor is applied in Layers. Each Layer is approximately 4% of the total tonnage of the Hull.

Tech Level. Armor Value AV= for a Layer is the Tech Level of the ship (potentially modified by Stage Effects).

Anti-Layers protect against specific threats.

ARMOR PROTECTS HULLS

Armor Layers are added inward from the Hull surface; Layers reduce available tonnage volume; they do not increase ship tonnage. Armor Layers (Table 09B) are homogeneous: they are all the same type. Anti-Layers (Table 09C) are modifications to Layers and substitute their values. Note Hull Structure and Configuration.

A. Determine number and tonnage of Armor Layers.

B. Identify Armor Type Code, AV and number of Layers on the FillForm.

C. Identify Anti-Layers (If any; potentially deferred)

D. Identify Coatings (if any; also potentially deferred).

A HULL ARMOR LAYERS

ID	Tons	1	2	3	4	5	6	7	8	9
A	100		4	8	12	16	20	24	28	32
B	200		8	16	24	32	40	48	56	64
C	300		12	24	36	48	60	72	84	96
D	400		16	32	48	64	80	96	112	128
E	500		20	40	60	80	100	120	140	160
F	600		24	48	72	96	120	144	168	192
G	700		28	56	84	112	140	168	196	224
H	800		32	64	96	128	160	192	224	256
J	900		36	72	108	144	180	216	252	288
K	1000		40	80	120	160	200	240	280	320
L	1100		44	88	132	176	220	264	308	352
M	1200		48	96	144	192	240	288	336	384
N	1300		52	104	156	208	260	312	364	416
P	1400		56	112	168	224	280	336	392	448
Q	1500		60	120	180	240	300	360	420	480
R	1600		64	128	192	256	320	384	448	512
S	1700		68	136	204	272	340	408	476	544
T	1800		72	144	216	288	360	432	504	576
U	1900		76	152	228	304	380	456	532	608
V	2000		80	160	240	320	400	480	560	640
W	2100		84	168	252	336	420	504	588	672
X	2200		88	176	264	352	440	528	616	704
Y	2300		92	184	276	368	460	552	644	736
Z	2400		96	192	288	384	480	576	672	768

Layer1 is integral to the Hull and has no tonnage cost.

Each layer after the first is 4% of Hull.

Additional Layers may be possible (conceivably down to about Layer24). Armor=Shell is half this tonnage.

On The FillForm. Specify **Armor Layers**. Note **Armor Type** Code and AV=. All Armor on a ship is identical (Pods and Subhulls may differ).

On The ShipSheet. Note **Anti-Layer Type Code** in each Layer Box. Each Layer Box may have (as desired) a different Anti-Layer.

B ARMOR LAYERS

Type	-Configuration-	TL=	AV=
A Plate	C B - U S - -	8	8 =TL
S Shell*	- - - S A L	9	5 =TL/2
P Polymer	C B - U S A -	10	5 =TL/2
O Organic	C B - U S A -	12	6 =TL/2
F FeNi	- - P U - - -	--	20 =20
C Charged	C B - U S - -	14	28 =TLx2

*Shell is half tonnage (on Table 09A)

C ANTI-LAYERS

Type	----- Effect -----	Multiplier=
B Anti-Blast	vs Blast Bullet Frag	x 10
E Anti-EMP	vs EMP	x 10
K Anti-Kinetic	vs Pen	x 10
R Anti-Rad	vs Rad	x 10
(applies to all)	vs Heat not Organic, Polymer	x100
(applies to all)	vs Pressure not Organic	x 10

D COATINGS

Type	----- Effect -----	Multiplier=
S Reflec	vs Burn or Heat	x100
T Ablat	vs Heat	x100
U Slick	vs Tractor	x100

One Coating is possible. It can be applied at no additional tonnage: it lies above the first layer of Armor.

ARMOR TECH LEVEL STAGE EFFECTS

TL	Stage	TL
Exp	Experimental	-3
Pro	Prototype	-2
Ear	Early	-1
Std	Standard	0
Imp	Improved	+1
Mod	Modified	+2
Adv	Advanced	+3
Ult	Ultimate	+4

For example, a 100-ton TL-10 Ship Configuration-S may have Armor= Plate, Shell, Polymer, Organic, or Charged (but not FeNi). Polymer Armor is Base TL=10. Layer1 imposes no additional tons; AV= (10/2=) 5. Layer2 (Table 09A) is 4 tons and AV=5.



10 Starship Drives

STARSHIP DRIVES

Power		8		8		19		14
	P		U		A		C	
	Power Plant		Fission		Anti-Matter		Collector	
Maneuver		9		8		6		
	M		G		R		H	
	Maneuver		Gravitic		Rocket		HEPlar	
Interstellar		9		17		20		9
	J		H		S		N	
	Jump		Hop		Skip		NAFAL	
Supplemental		5		8		9		
	B		H		F+			
	Battery		Fuel Cell		Fusion Plus			

SELECTING DRIVES

A ship may require:

An **Interplanetary Drive** (M, G, R, H) for travel within a star system and to and from world surfaces. Default is M.

An **Interstellar Drive** (N, J, H, S) to allow travel between star systems. The default is J.

A **Power System** (A, C, U, P) to provide energy for operations. The default is P. In some cases, an alternative Power Source (B, H, F+) may be selected.

W. Tech Level Availability.

X. Drive Efficiency

Y. Drive Cost.

Z1 and Z2. Drive Potential.

Drive Efficiency

Drives (and Power Systems) produce full (100 percent) Potential (Table 11Y) at base TL. Other TLs may produce other Potentials based on Drive Efficiency. Efficiency also affects fuel consumption (if applicable). Fuel is the multiplier for fuel required (or time required to charge for Collectors).

Tech Level Limits On Availability

Tech Level Availability shows the Drive Potential available for a Standard drive at Base Tech Level. For example, G-Drive Potential-1 is available at TL-8, and G-Drive Potential 2-3-4 is available at TL-9. Theoretically, a TL-5 planetary economy can produce an Experimental G-Drive.

W DRIVE TECH LEVEL AVAILABILITY

Drive	TL=	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
M Maneuver		-	-	1	3	5	7	9	-	-	-	-	-	-	-	-
G Gravitic		-	1	4	7	9	-	-	-	-	-	-	-	-	-	-
R Rocket		1	4	7	9	-	-	-	-	-	-	-	-	-	-	-
H HEPlar		-	1	3	5	7	9	-	-	-	-	-	-	-	-	-
N NAFAL		-	-	1	4	7	9	-	-	-	-	-	-	-	-	-
J Jump		-	-	1	-	2	3	4	5	6	7	8	9	-	-	-
H Hop		-	-	-	-	-	-	-	-	-	-	1	-	2	3	4
S Skip		-	-	-	-	-	-	-	-	-	-	-	-	-	1	-
P P-Plant		-	1	2	3	4	5	6	7	8	9	-	-	-	-	-
A Anti-Matter		-	-	-	-	-	-	-	-	-	-	-	-	1	2	3
U Fission		1	2	3	4	5	6	7	8	9	-	-	-	-	-	-
C Collector		-	-	-	-	-	-	-	1	2	3	4	5	6	7	8

Tech Level Availability shows the Drive Potential available for a Standard drive at its Base Tech Level. For example, G-Drive Potential-1 (capable of 1G) is available at TL-8, and G-Drive Potential-4 (as well as Potential-3 and Potential-2) is available at TL-9.

X DRIVE TECH LEVEL STAGE EFFECTS

Stage	TL	Cost	Eff	Fuel	Tons
Exp Experimental	-3	x10	50%	2.0	x3
Pro Prototype	-2	x5	80%	1.2	x2
Ear Early	-1	x2	90%	1.1	x1
Std Standard	0	x1	100%	1.0	x1
Bas Basic	0	/2	90%	1.1	x1
Alt Alternate	0	x1	100%	1.0	x1
Imp Improved	+1	x1	110%	0.9	x1
Gen Generic	+1	/2	90%	1.1	x1
Mod Modified	+2	x1	110%	0.9	/2
Adv Advanced	+3	x2	120%	0.8	/3
Ult Ultimate	+4	x3	130%	0.7	/4

TL Stage Effects (Cost, QREBS) apply.

Efficiency modifies the EP Energy Point output of a Drive. Standard Drive-C has 300 EP; Early Drive-C outputs 150 EP; Advanced Drive-C outputs 360 EP.



Ships can be equipped with Interplanetary and Interstellar drives, and power systems.

The Drive Costs Table determines drive tonnage.

Starship Drives 11

		In-System				Interstellar				Power Systems			
Y	DRIVE TONS												
		M	G	R	H	J	H	S	N	A	C	U	P
Drive	EP	Man	Grav	Rocket	HEPlar	Jump	Hop	Skip	NAFAL	AM	Collect	Fission	Power
A	100	2	9	2	1	10	15	10	2	31	20	15	4
B	200	3	18	4	2	15	20	15	4	32	30	20	7
C	300	5	27	6	3	20	25	20	6	33	40	25	10
D	400	7	36	8	4	25	30	25	8	34	50	30	13
E	500	9	45	10	5	30	35	30	10	35	60	35	16
F	600	11	54	12	6	35	40	35	12	36	70	40	19
G	700	13	63	14	7	40	45	40	14	37	80	45	22
H	800	15	72	16	8	45	50	45	16	38	90	50	25
J	900	17	81	18	9	50	55	50	18	39	100	55	28
K	1000	19	90	20	10	55	60	55	20	40	110	60	31
L	1100	21	99	22	11	60	65	60	22	41	120	65	34
M	1200	23	108	24	12	65	70	65	24	42	130	70	37
N	1300	25	117	26	13	70	75	70	26	44	140	75	40
P	1400	27	126	28	14	75	80	75	28	46	150	80	43
Q	1500	29	135	30	15	80	85	80	30	48	160	85	46
R	1600	31	144	32	16	85	90	85	32	50	170	90	49
S	1700	33	153	34	17	90	95	90	34	52	180	95	52
T	1800	35	162	36	18	95	100	95	36	54	190	100	55
U	1900	37	171	38	19	100	105	100	38	56	200	105	58
V	2000	39	180	40	20	105	110	105	40	68	210	110	61
W	2100	41	189	42	21	110	115	110	42	60	220	115	64
X	2200	43	198	44	22	115	120	115	44	62	230	120	67
Y	2300	45	207	46	23	120	125	120	46	64	240	125	70
Z	2400	47	216	48	24	125	130	125	48	66	250	130	73
N2	2600	50	234	52	26	140	150	140	52	88	280	150	80
P2	2800	54	252	56	28	150	160	150	56	92	300	160	86
Q2	3000	58	270	60	30	160	170	160	60	96	320	170	92
R2	3200	62	288	64	32	170	180	170	64	100	340	180	98
S2	3400	66	306	68	34	180	190	180	68	104	360	190	104
T2	3600	70	324	72	36	190	200	190	72	108	380	200	110
U2	3800	74	342	76	38	200	210	200	76	112	400	210	116
V2	4000	78	360	80	40	210	220	210	80	116	420	220	122
W2	4200	82	378	84	42	220	230	220	84	120	440	230	128
X2	4400	86	396	88	44	230	240	230	88	124	460	240	134
Y2	4600	90	414	92	46	240	250	240	92	128	480	250	140
Z2	4800	94	432	96	48	250	260	250	96	132	500	260	146
MCr/ton=		2.0	0.5	0.5	1.0	1.0	2.0	3.0	1.0	2.0	0.5	1.5	1.0

No drive may be smaller than the Drive-A of the class, even after modifications like Stage Effects.

Example: A TL12 200-ton Hull-B can select a TL12 Jump Drive-D (Table 11Y above) at a cost of 25 tons and MCr25.

At TL12, the maximum Jump a standard drive can produce (Table 10W) is Jump-3 based on standard TL. Jump Drive-D in Hull-B can produce (Table 12Z1) Jump-4 (but 10W constrains it to Jump-3). The Hull-B Jump-D combination, assuming a Power Plant with equal potential, requires (Table 13) $3 \times 10\% \times 200$ tons = 60 tons of fuel for a Jump-3.

A society at TL 16 cannot produce standard Hop Drives. It can produce a Hop-A for installation in a Hull-B as an Experimental Hop-2 (at 50% efficiency, round down to Hop-1) with various tonnage (x3), cost (x10), and fuel (x2) handicaps, and it would require a Jump Field based on Jump Plates. Nevertheless, the result is an improvement over most other options.



12 Starship Drive Potential

Z1 DRIVE POTENTIAL

Determine Drive Potential (in table body)
for a Drive (left column) installed in a Hull (top row).

	Hull=	A	B	C	D	E	F	G	H	J	K	L	M	N	P	Q	R	S	T	U	V	W	X	Y	Z		
Drive=	A	2	1	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	A	
	B	4	2	1	1	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	B	
	C	6	3	2	1	1	1	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	C	
	D	8	4	2	2	1	1	1	1	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	no	D	
	E	9	5	3	2	2	1	1	1	1	1	no	no	no	no	no	no	no	no	no	no	no	no	no	no	E	
	F	9	6	4	3	2	2	1	1	1	1	1	1	no	no	no	no	no	no	no	no	no	no	no	no	F	
	G	9	7	4	3	2	2	2	1	1	1	1	1	1	1	no	no	no	no	no	no	no	no	no	no	G	
	H	9	8	5	4	3	2	2	2	1	1	1	1	1	1	1	1	no	no	no	no	no	no	no	no	H	
	J	9	9	6	4	3	3	2	2	2	1	1	1	1	1	1	1	1	1	1	no	no	no	no	no	J	
	K	9	9	6	5	4	3	2	2	2	2	1	1	1	1	1	1	1	1	1	1	no	no	no	no	K	
	L	9	9	7	5	4	3	3	2	2	2	2	1	1	1	1	1	1	1	1	1	1	1	no	no	L	
	M	9	9	8	6	4	4	3	3	2	2	2	2	1	1	1	1	1	1	1	1	1	1	1	1	M	
Drive=	N	9	9	8	6	5	4	3	3	2	2	2	2	2	1	1	1	1	1	1	1	1	1	1	1	N	
	P	9	9	9	7	5	4	4	3	3	2	2	2	2	2	1	1	1	1	1	1	1	1	1	1	P	
	Q	9	9	9	7	6	5	4	3	3	3	2	2	2	2	2	1	1	1	1	1	1	1	1	1	Q	
	R	9	9	9	8	6	5	4	4	3	3	2	2	2	2	2	2	1	1	1	1	1	1	1	1	R	
	S	9	9	9	8	6	5	4	4	3	3	3	2	2	2	2	2	2	1	1	1	1	1	1	1	S	
	T	9	9	9	9	7	6	5	4	4	3	3	3	2	2	2	2	2	2	1	1	1	1	1	1	T	
	U	9	9	9	9	7	6	5	4	4	3	3	3	2	2	2	2	2	2	2	1	1	1	1	1	U	
	V	9	9	9	9	8	6	5	5	4	4	3	3	3	2	2	2	2	2	2	2	1	1	1	1	V	
	W	9	9	9	9	8	7	6	5	4	4	3	3	3	3	2	2	2	2	2	2	2	1	1	1	W	
	X	9	9	9	9	8	7	6	5	4	4	4	3	3	3	2	2	2	2	2	2	2	2	1	1	X	
	Y	9	9	9	9	9	7	6	5	5	4	4	3	3	3	3	2	2	2	2	2	2	2	2	1	Y	
	Z	9	9	9	9	9	8	6	6	5	4	4	4	3	3	3	3	2	2	2	2	2	2	2	2	Z	
Drive=	N2	9	9	9	9	9	8	7	6	5	5	4	4	4	3	3	3	3	2	2	2	2	2	2	2	N2	
	P2	9	9	9	9	9	9	8	7	6	5	5	4	4	4	3	3	3	3	2	2	2	2	2	2	P2	
	Q2	9	9	9	9	9	9	8	7	6	6	5	5	4	4	4	3	3	3	3	3	2	2	2	2	Q2	
	R2	9	9	9	9	9	9	8	7	6	5	5	4	4	4	4	3	3	3	3	3	3	2	2	2	R2	
	S2	9	9	9	9	9	9	9	8	7	6	6	5	5	4	4	4	4	3	3	3	3	3	2	2	S2	
	T2	9	9	9	9	9	9	9	9	8	7	6	6	5	5	4	4	4	4	3	3	3	3	3	3	3	T2
	U2	9	9	9	9	9	9	9	9	8	7	6	6	5	5	5	4	4	4	4	3	3	3	3	3	U2	
	V2	9	9	9	9	9	9	9	9	8	8	7	6	6	5	5	5	4	4	4	4	3	3	3	3	V2	
	W2	9	9	9	9	9	9	9	9	9	8	7	7	6	6	5	5	4	4	4	4	4	3	3	3	W2	
	X2	9	9	9	9	9	9	9	9	9	8	8	7	6	6	5	5	5	4	4	4	4	4	3	3	X2	
Y2	9	9	9	9	9	9	9	9	9	9	8	7	7	6	6	5	5	5	5	4	4	4	4	3	Y2		

no= not possible. For example, Maneuver Drive-T (row T) in Hull-E (column E) produces Maneuver Drive Potential = 7.

Table Z1 shows Jump Drive-K in Hull-K produces Potential 2 (indeed, any Drive Letter = Hull Letter produces Potential-2).

Z2 DRIVE POTENTIAL

Determine the Drive (in table body) for a specific Potential (left column) for a Hull (top row).

Potential=	Hull=	A	B	C	D	E	F	G	H	J	K	L	M	N	P	Q	R	S	T	U	V	W	X	Y	Z
1	A	A	B	B	C	C	D	D	E	E	F	F	G	G	H	H	J	J	K	K	L	L	M	M	1
2	A	B	C	D	E	F	G	H	J	K	L	M	N	P	Q	R	S	T	U	V	W	X	Y	Z	2
3	B	C	E	F	H	J	L	M	P	Q	S	T	V	W	Y	Z	N2	P2	Q2	Q2	R2	S2	T2	T2	3
4	B	D	F	H	K	M	P	R	T	V	X	Z	N2	P2	Q2	R2	S2	T2	U2	V2	W2	X2	Y2	Z2	4
5	C	E	H	K	N	Q	T	V	Y	N2	P2	Q2	S2	T2	U2	V2	X2	Y2	Z2	no	no	no	no	no	5
6	C	F	J	M	Q	T	W	Z	P2	Q2	S2	T2	V2	W2	Y2	Z2	no	no	no	no	no	no	no	no	6
7	C	G	L	P	T	W	N2	P2	R2	T2	V2	W2	Y2	no	no	no	no	no	no	no	no	no	no	no	7
8	D	H	M	R	V	Z	P2	R2	T2	V2	X2	Z2	no	no	no	no	no	no	no	no	no	no	no	no	8
9	E	J	P	T	Y	P2	R2	T2	W2	Y2	no	no	no	no	no	no	no	no	no	no	no	no	no	no	9

no= not possible. For example, achieving Jump-6 (row 6) in a 1000-ton Hull-K (column K) requires Jump-Drive-Q2.



FUEL REQUIREMENTS

Determine drive and power fuel needs. Allocate fuel for (at least) one month of operations and (if equipped) one interstellar journey.

F. **Allocate Fittings** as required.

Starship Fuel 13

		 P	 U	 A	 C	 G Gravitic (Near World Drive) internally self-fuelled; no fuel required)	
		Power Plant	Fission Plant	Anti- Matter	Collector		
Fuel		Liquid Hydrogen in Tons Ton=Cr500	Radioactives in Rods Rod= KCr40	Anti-Matter in Slugs Slug= KCr6	Exotic Particles in Charges		
 M Maneuver		included in Operations	not possible	included in Operations	not possible	gravity-based interplanetary drive.	M
 R Rocket		uses Rocket Fuel	uses Rocket Fuel	not possible	not possible	thrust-based interplanetary drive.	R
 H HEPlAR		1 Burn***= P x Hull /100	1 Burn***= P x Hull /100	not possible	not possible	improved thrust-based interplanetary drive.	H
 J Jump		Tons= P x Hull /10	not possible	Slugs= P x Hull /100	one full charge	interstellar drive. fuel in addition to Operations.	J
 H Hop		Tons= P x Hull /10	not possible	Slugs= P x Hull /100	one full charge	interstellar drive. fuel in addition to Operations.	H
 S Skip		Tons= P x Hull /10	not possible	Slugs= P x Hull /100	one full charge	interstellar drive. fuel in addition to Operations.	S
 N NAFAL		Tons= P x Hull /100 for 1 month	included in Operations*	included in Operations*	not possible	sublight interstellar drive. *If NAFAL in use, fuel use x2.	N
Y Operations		Tons= P x Hull /100 for 1 month**	Rods= P x Hull /100 for 10 years**	Slugs= P x Hull /100 for 1 year**	not possible	general energy requirements for life support and activity.	Y

** Decentralized power; if centralized, fuel use x2. P= Drive Potential Used. H= Hull Tonnage.

*** In addition to Rocket Fuel thrust requirements.

F Fuel Costs and Fittings

Item	Cost	TL	Comment	
Raw Fuel (Gas)	*	0	Unrefined Hydrogen Gas= Skimmed from gas giants.	Hydrogen
Raw Fuel (Water)	*	0	Water= Pumped from oceans or lakes.	
Raw Fuel (Ice)	*	0	Ice= Gathered from ice caps, asteroids.	
Raw Fuel (Gas)	Cr100 / ton	2	at Starport ABCD	
Refined Fuel (Gas)	Cr500 / ton	3	at Starport AB	Power Plant
S Fuel Scoop	KCr100 1 ton	8	Intakes 100 tons of gas / hour. Intakes gas into fuel tanks.	
N Fuel Intake	KCr100 1 ton	8	Intakes 40 tons of water/ hour. Pumps liquid (hull ports / hoses).	
B Fuel Bin	KCr100 1 ton	8	Intakes 20 tons of ice/ hour. Accepts ice solids.	
P +Fuel Purifier	MCr1 1 ton	8	Converts raw fuel to refined fuel at the rate of the intake.	U
X Transfer Pump	MCr1 1 ton	10	Transfers 100 tons of Drop Tank fuel to P-Plant in seconds.	
U Fuel Rod	KCr400 / set	8	at Starports AB TL 8+. Set= 10 Rods. Store up to 200 Rods per ton.	C
C Canopy	=half Collector cost	14	at Starports AB TL 14+. Replaces old Collector Canopy.	
A AM Fuel Slugs	KCr60 / set	18	at Starports AB TL 18+. Set= 10 Slugs. Store up to 1,000 Slugs per ton.	AM

AM Slug Production. An AM Plant can operate "in reverse" to produce AM Slugs. Supported by a Power Plant of equal potential, an AM Plant in Production Mode can produce 1 AM slug per 100 EP per day.



Starship Sensors2 15

Sensors gather information about potential threats.

A SENSORS

	Type	TL	Mount	S=	R=	MCR	Passive	Active (Principle)	Sensors + Trade	
Space Range Sensors Com/Visual	C Communicator	8	Surf	7		1.0	Passive	Active (Electronics)	Electronics. Comms.	C
	H Holovisor	18	Surf	7		1.0	Passive		Photonics	H
	T Scope	9	Surf	7		1.0	Passive		Photonics	T
	V Visor	14	Surf	7		1.0	Passive		Photonics	V
	W CommPlus	15	Surf	7		1.0	Passive	Active (Gravitics)	Gravitics	W
Space Range Sensors Space	E EMS	12	Surf	7		1.0	Passive	Active (Electronics)	Electronics	E
	G Grav Sensor	13	Surf	7		1.0	Passive		Gravitics	G
	N Neutrino Detector	10	Surf	7		1.0	Passive		Gravitics	N
	R Radar	9	Surf	7		1.0	Passive	Active (Electronics)	Electronics	R
	S Scanner	19	Surf	7		1.0	Passive	Active (Electronics)	Electronics	S
World Range Sensors World	A Activity Sensor	11	Surf		7	0.1	Passive		Electronics	A
	B Deep Radar	9	Surf		7	0.1		Active (Electronics)	Electronics	B
	D Densitometer	10	Surf		7	0.1	Passive		Gravitics	D
	L Life Detector	10	Surf		7	0.1	Passive		Magnetics	L
	M Mass Sensor	8	Surf		7	0.1	Passive		Gravitics	M
World Range Sensors Specialized	U Radiation Sensor	8	Surf		7	0.1	Passive		Electronics	U
	F Field Sensor	12	Surf		7	0.1	Passive		Electronics (Magnetics)	F
	K Analyzer/ Sniffer	9	Surf		7	0.1	Passive		Electronics (Biologics)	K
	P Proximeter	10	Surf		7	0.1		Active (Electronics)	Electronics	P
	Y Sound Sensor	7	Surf		7	0.1	Passive		Electronics	Y
Jam	J Jammer	8	Surf		7	1.0		Active (Electronics)	Electronics (Gravitics)	J
	Q Stealth Mask	12	Surf		7	1.0	Passive		Polymers	Q

C SENSOR MOUNTS

Mount	Type	Tons	Mod	Skill	MCR
Surf	Surface	0	0	Sensor	1.0
Ant	Antenna	1	+1	Sensor	0.5
T1	Single Turret	1	-2	Sensor	0.2
T2	Dual Turret	1	-1	Sensor	0.5
T3	Triple Turret	1	0	Sensor	1.0
T4	Quad Turret	1	+1	Sensor	1.5
B1	Barbette	3	+2	Sensor	3.0
B2	Dual Barbette	5	+3	Sensor	4.0
BAnt	Big Antenna	10	+5	Sensor	2.0
Bay	Bay	50	+5	Sensor	5.0
LBay	Large Bay	100	+8	Sensor	10.0
M	Main	200	+10	Sensor	20.0
Ext	Extendable	+2	+3	Sensor	1.0
De	Deployable	+2		Sensor	3.0

Deployable/Extendable. In addition to Turret cost

B TECH LEVEL STAGE EFFECTS

TL	Stage	TL	Cost	Mod
Exp	Experimental	-3	x10	-3
Pro	Prototype	-2	x5	-2
Ear	Early	-1	x2	-1
Std	Standard	0	x1	0
Bas	Basic	0	/2	0
Alt	Alternate	0	x1	0
Imp	Improved	+1	x1	+1
Gen	Generic	+1	/2	0
Mod	Modified	+2	/2	+2
Adv	Advanced	+3	x2	+3
Ult	Ultimate	+4	x3	+4

Applies to Sensor. (not Mount).

INSTALLING SENSORS

- Select a **Sensor**.
- TL Stage Effects.**
- Select a **Mount**.
- Space Range**, or
- World Range**.
- Control Panel**.
- Note cost / tonnage.
One Sensor Mount per **HardPoint**; mount is at least 1 ton (round up).
One Sensor Mount per **FirmPoint**; must be less than 1 ton.
- Sensor Surface Mounts** do not require **HardPoints** or **FirmPoints**.

D SPACE RANGE EFFECTS

S=	Range	TL	Tons	Cost
0	BR Boarding	-3	/4	/4
2	FR Fighter Range	-2	/3	/3
5	SR Short Range	-1	/2	/2
7	AR Attack Range	0	x1	x1
9	LR Long Range	+1	x2	x3
12	DS Deep Space	+2	x3	x5

For Sensors using S= Space Range.
Convert S=R-5. Applies to Mount.

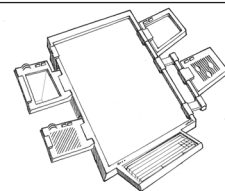
E WORLD RANGE EFFECTS

S=	R=	Range	TL	Tons	Cost
0	5	VL Vlong 1000 m	-2	/3	/3
1	6	D Distant 5 km	-1	/2	/2
2	7	Vd VDistant 50 km	0	x1	x1
3	8	Or Orbit 500 km	+1	x2	x3
4	9	Fo Far 5,000 km	+2	x3	x5
5	10	G Geo 50,000 km	+3	x4	x6

For Sensors using R= World Range.
Convert R=S+5. Applies to Mount

F CONTROLS

Install Control Panels.



One per Sensor.



Starship Weapons2 17

Weapons project power to dominate or defend against potential threats.

A WEAPONS

	Type	TL	Mount	S=	R=	MCR	Principle	Skills plus Knowledge:			
Beams	J Mining Laser	8	Turret		7		0.5 Electronic.	Turret.	Bay.	Main.	J
	K Pulse Laser	9	Turret		7		0.3 Electronic.	Turret.	Bay.	Main.	K
	L Beam Laser	10	Turret		7		0.5 Electronic.	Turret.	Bay.	Main.	L
	P Plasma Gun	11	Barbette		7		1.0 Electronic. Gravitic.	Turret.	Bay.	Main.	P
	F Fusion Gun	12	Barbette		7		1.5 Electronic. Gravitic.	Turret.	Bay.	Main.	F
Missiles	B Slug Thrower	9	Turret		7		0.2 Electronic.	Turret.	Bay.	Main.	B
	V Salvo Rack	10	Bay		7		10.0 Electronic. Magnetic.		Bay.	Main.	V
	R Rail Gun	12	Bay	7			12.0 Electronic. Magnetic.	Ortillery.			R
	M Missile	7	Turret	7			2.0 Electronic.	Turret.	Bay.	Main.	M
	N KK Missile	10	Bay	7			3.0 Electronic.		Bay.	Main.	N
Exotics	X AM Missile	20	Barbette	7			5.0 Electronic.	Turret.	Bay.	Main.	X
	T Jump Damper	14	Barbette		7		15.0 Electronic.	Turret.	Bay.	Main.	T
	U Tractor/Pressor	16	Barbette		7		5.0 Electronic.	Turret.	Bay.	Main.	U
	H Inducer	17	Turret		7		1.0 Electronic.	Turret.	Bay.	Main.	H
	W Disruptor	18	Barbette	7			15.0 Electronic. Gravitic.	Turret.	Bay.	Main.	W
Special	E Stasis	21	Turret	7			5.0 Electronic.	Turret.	Bay.	Main.	E
	D DataCaster	10	Turret		7		1.0 Electronic.	Turret.	Bay.	Main.	D
	S SandCaster	9	Turret		7		0.1 Electronic.	Turret.	Bay.	Main.	S
	Q Ortillery	12	Bay		7		15.0 Electronic.	Ortillery.			Q
	C CommCaster	8	Turret	7			5.0 Electronic.	Turret.	Bay.	Main.	C
LR	Y Hybrid S-L-M	10	Turret	7*	7*		1.0 Electronic.	Turret.	Bay.	Main.	Y
	A Particle Accel	11	Barbette	7*	7*		2.5 Electronic. Magnetic.	Turret.	Bay.	Main.	A
	G Meson Gun	13	Main	7			5.0 Electronic. Gravitic.			Main.	G

C WEAPON MOUNTS

Mount	Type	Tons	Mod	Hits (D)	MCR
T1	Single Turret	1	-2	1	0.2
T2	Dual Turret	1	-1	2	0.5
T3	Triple Turret	1	0	3	1.0
T4	Quad Turret	1	+1	4	1.5
B1	Barbette	3	+2	5	3.0
B2	Dual Barbette	5	+3	10	4.0
Bay	Bay	50	+5	20	5.0
LBay	Large Bay	100	+8	30	10.0
M	Main	200	+10	100	20.0
Ext	Extendable	+2	+3		1.0
De	Deployable	+2			3.0

Deployable/Extendable. In addition to Turret cost.

B TECH LEVEL STAGE EFFECTS

TL	Stage	TL	Cost	Mod
Exp	Experimental	-3	x10	-3
Pro	Prototype	-2	x5	-2
Ear	Early	-1	x2	-1
Std	Standard	0	x1	0
Bas	Basic	0	/2	0
Alt	Alternate	0	x1	0
Imp	Improved	+1	x1	+1
Gen	Generic	+1	/2	0
Mod	Modified	+2	/2	+2
Adv	Advanced	+3	x2	+3
Ult	Ultimate	+4	x3	+4

Applies to Weapon. (not Mount).

INSTALLING WEAPONS

- Select a **Weapon**.
- TL Stage Effects**.
- Select a **Mount**.
- Space Range**, or
- World Range**.
- Control Panel**.
- Note cost / tonnage.
One Weapon Mount per **HardPoint**; mount is at least 1 ton (round up).
One Weapon Mount per **Firmpoint**; must be less than 1 ton.

D SPACE RANGE EFFECTS

S=	Range	TL	Tons	Cost
0	BR Boarding	-3	/4	/4
2	FR Fighter Range	-2	/3	/3
5	SR Short Range	-1	/2	/2
7	AR Attack Range	0	x1	x1
9	LR Long Range	+1	x2	x3
12	DS Deep Space	+2	x3	x5

For Sensors using S= Space Range.
Convert S=R-5. Applies to Mount.

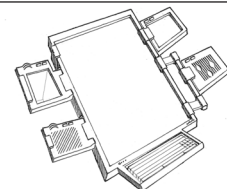
E WORLD RANGE EFFECTS

S=	R=	Range	TL	Tons	Cost
0	5	VL Vlong 1000 m	-2	/3	/3
1	6	D Distant 5 km	-1	/2	/2
2	7	Vd VDistant 50 km	0	x1	x1
3	8	Or Orbit 500 km	+1	x2	x3
4	9	Fo Far 5,000 km	+2	x3	x5
5	10	G Geo 50,000 km	+3	x4	x6

For Sensors using R= World Range.
Convert R=S+5. Applies to Mount

F CONTROLS

Install Control Panels.



One per Sensor.



Starship Defenses2 19

Defenses reduce or eliminate threats.

A DEFENSES

Type	TL	Mount	S=	R=	MCR	Principle.	Skill + Knowledge:	
J Mining Laser	8	Turret		7	0.5			J
K Pulse Laser	9	Turret		7	0.3			K
L Beam Laser	10	Turret		7	0.5	Installed As A Weapon.		L
P Plasma Gun	11	Barbette		7	1.0			P
F Fusion Gun	12	Barbette		7	1.5			F
B Slug Thrower	9	Turret		7	0.2			B
V Salvo Rack	10	Bay		7	10.0	Installed As A Weapon.		V
S SandCaster	9	Turret		7	0.1			S
Y Hybrid S-L-M	10	Turret	7*	7*	1.0	Installed As A Weapon.		Y
N Nuclear Damper	12	Bo		7	1.0	Electronic.	Turret. Bay. Main.	N
G Meson Screen	13	Bo		7	3.0	Electronic.	Turret. Bay. Main.	G
R Proton Screen	19	Bo		7	1.0	Electronic.	Turret. Bay. Main.	R
Q Mag Scrambler	14	Bo		7	1.0	Magnetic	Screens.	Q
J Grav Scrambler	17	Bo		7	2.0	Gravitic.	Screens.	J
E Elec Scrambler	12	Bo		7	2.0	Electronic.	Screens.	E
T Black Globe	16	Bo		7	10.0	Electronic.	Screens.	T
U White Globe	20	Bo		7	10.0	Electronic.	Screens.	U
V Silver Globe	22	Bo		7	10.0	Electronic.	Screens.	V
W Stasis Globe	24	Bo		7	10.0	Electronic.	Screens.	W
J Jammer	8	Surf		7	1.0	Installed As A Sensor.		J
Q Stealth Mask	12	Surf	7		1.0			Q

C DEFENSE MOUNTS

Mount	Type	Tons	Mod	MCR
Bo	Bolt-In	3	3	3.0
T1	Single Turret	1	1	0.2
T2	Dual Turret	1	2	0.5
T3	Triple Turret	1	3	1.0
T4	Quad Turret	1	4	1.5
B1	Barbette	3	1	3.0
B2	Dual Barbette	5	2	4.0
Bay	Bay	50	1	5.0
LBay	Large Bay	100	1	10.0
M	Main	200	1	20.0
Ext	Extendable	+2		1.0
De	Deployable	+2		3.0

Deployable/Extendable. In addition to Turret cost.

B TECH LEVEL STAGE EFFECTS

TL	Stage	TL	Cost	Mod
Exp	Experimental	-3	x10	-3
Pro	Prototype	-2	x5	-2
Ear	Early	-1	x2	-1
Std	Standard	0	x1	0
Bas	Basic	0	/2	0
Alt	Alternate	0	x1	0
Imp	Improved	+1	x1	+1
Gen	Generic	+1	/2	0
Mod	Modified	+2	/2	+2
Adv	Advanced	+3	x2	+3
Ult	Ultimate	+4	x3	+4

Applies to Defense (not Mount).

INSTALLING DEFENSES

- Select a **Defense**.
- TL Stage Effects.**
- Select a **Mount**.
- Space Range**, or
- World Range**.
- Control Panel**.
- Note cost / tonnage.
- One Defense Mount per **HardPoint**; mount is at least 1 ton (round up).
- One Weapon Mount per **Firmpoint**; must be less than 1 ton.

D SPACE RANGE EFFECTS

S=	Range	TL	Tons	Cost
0	BR Boarding	-3	/4	/4
2	FR Fighter Range	-2	/3	/3
5	SR Short Range	-1	/2	/2
7	AR Attack Range	0	x1	x1
9	LR Long Range	+1	x2	x3
12	DS Deep Space	+2	x3	x5

For Defenses using S= Space Range.
Convert S=R-5. Applies to Mount.

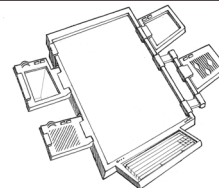
E WORLD RANGE EFFECTS

S=	R=	Range	TL	Tons	Cost
0	5	VL Vlong 1000 m	-2	/3	/3
1	6	D Distant 5 km	-1	/2	/2
2	7	Vd VDistant 50 km	0	x1	x1
3	8	Or Orbit 500 km	+1	x2	x3
4	9	Fo Far 5,000 km	+2	x3	x5
5	10	G Geo 50,000 km	+3	x4	x6

For Defenses using R= World Range.
Convert R=S+5. Applies to Mount.

F CONTROLS

Install Control Panels.



One per Sensor.



Starship Operations 21

OPERATIONS

- Determine Operational fittings.
- A. Determine Ship's Troops.
- B. Determine Specialists.
- C. Determine Life Support.
- V. Assign Vehicles
- Q Assign Small Craft.
- F. Assign Fuel Tankage.
- G. Assign Vehicle Storage.

SHIP'S TROOPS BARRACKS

Assign (as needed) Ship's Troops.

A TROOP ACCOMMODATIONS

Quarters	Tons	MCr
S Squad Barracks (5)	4	0.2
P Platoon Barracks (22)	18	1.0
C Company Barracks (70)	60	3.0

(N) = typical number of troops. The actual number assigned may vary.

Typical unit sizes are:

Squad. Five troops with ranks R3 R2 R2 R1 R1.

Platoon. Four squads, plus an officer O2 or O1 and a sergeant R4.

Company. Three Platoons, plus a command section O3, O2, R5, R3.

Each Ship's Troops Squad is assigned as the crew for one Ship's Weapon or Defense (per Squad).

F FUEL STORAGE

Fuel	Power System	Capacity
Liquid Hydrogen	Power Plant	one ton holds one ton
Anti-Matter	Anti-Matter Plant	one ton holds 1000 Slugs
Radioactives	Fission Plant	one ton holds 200 Rods
Particles	Collector	energy is stored in the Collector Canopy.

There is no cost associated with Fuel Storage systems.

G VEHICLE HULL CONNECTORS

Type	Function	TL	Tons	Configuration	MCr	Comment
B Bracket	Exterior Vehicle	7	1	C B P U	1	1 per Vehicle (maximum 10 tons).
S Streamlined Bracket	Exterior Vehicle	7	1	S A L	2	1 per Vehicle (maximum 10 tons).
N Hull Niche	Exterior Vehicle	8	varies	C B P U S A L	1.5	Tons= Half Vehicle.
H Dedicated Hangar	Interior Vehicle	7	varies	C B P U S A L		Tons= Vehicle +10
C Interior/ Hold	Vehicle Storage	7	varies	C B P U S A L		Tons= Vehicle + 1.
G Grapple	Exterior Pod	9	1	C B P U S A L	1	1 pair per 35 tons
R Subhull Grapple	Exterior Subhull	9	1	C B P U S A L	1	Subhull requires 3 pairs per 100 tons
E Connector	Permanent	9	0.5	C B P U S A L	1	Replaces Grapple; no disconnect.

Grapple Pairs located one per parent ship and one per carried Pod or Subhull. Brackets required only on the parent ship.

SPECIALISTS

Assign (as needed) specialists.

B SPECIALISTS

Assign Specialty Consoles:

Political Officer	Broker
Cultural Officer	Trader
Liaison Specialist	Linguist
Contact Specialist	Negotiator
Translator	Advocate
Sensop Specialist	Diplomat

consistent with the ship mission.

LIFE SUPPORT

Assign tonnage for Life Support. Life Support is a job responsibility assigned to a Steward. Each Life Support mechanism has a Control Panel.

C LIFE SUPPORT

Type	Person-Days	Tons	MCr
H Short Term	40	0	0
S Standard	300	1	1.0
L Long-Term	1200	2	2.0
U Luxury	30	1	1.0
A Adaptable	30	1	1.0

Adaptable required for non-human environments. Luxury required for High Passage staterooms.

VEHICLES

Assign vehicles to the ship for mobility, exploration, and mission support.

Vehicles carried by a ship occupy tonnage. Specialized connectors allow more efficient storage.

Install an appropriate Vehicle Hull Connector for each Vehicle. Each Vehicle Hull Connector has a Control Panel.

V VEHICLE		Tons	MCr
Air/Raft	R	4	0.06
Flyer	F	10	0.35
ATV-W	A	5	0.16
Car	C	2	0.03
Truck-W	T	4	0.05
Vehicle	V	5	0.06
ATV-T	A	7	0.32
Boat	B	10	0.10
Submarine	U	100	1.00

Q SMALL CRAFT		Tons	MCr
Lifepod	A	10	2.0
Fighter	F	10	9.0
Gig	G	20	2.0
Launch	L	20	7.0
Ship's Boat	B	30	6.0
Pinnacle	N	40	8.0
Cutter	C	50	13.0
Cutter Module	M	30	6.0
Shuttle	S	95	11.0
Pod	P	<100	
Drone	D	varies	

Fuel Tankage

Evaluate ship fuel requirements and assign appropriate fuel storage.



22 Starship Controls

STARSHIP CONTROLS

Consoles are supervised by one or more Ship's Computers.

A. Note available Control Panels.

B. Install Consoles and associate each with one or more Control Panel. Note the three types of Console (Control, Operating, and Workstation).

C. Select a Staffing Level and calculate the required number of crew.

D. Install Ship's Computer(s), and (if necessary) note a Master Computer.

CONTROL PANELS

Every mechanism and function has a Control Panel.

A **mechanism** is any of the drives, sensors, weapons, defenses, or installations that equip a ship. For any device greater than 35 tons, each 35 tons is treated as a separate mechanism.

FillForm Control Panels. Assign a Control Panel to each mechanism on the FillForm (or 1 per 35 tons if the mechanism is more than 35 tons). A simple penciled stroke is sufficient. Shaded Con boxes do not require Control Panels.

A ASSIGN CONSOLE FUNCTIONS

Skill	Function
Astrogation	Plots courses.
Pilot	Flight controls.
Security	Internal barriers.
Counselling	Crew mental health.
Freight	Loading/Unloading.
Steward	Passenger/Life Support.

CONSOLES

Identify each console by a single Ehex identifier 0 through Z (above 33, assign double digit Ehex 00 to ZZ).

Connections. A Console controls one or more Control Panels; assign each CP to a Console (write Console ID for each CP in the Con Column).

Multi-Tasking. Any number of Control Panels, Mechanisms, or Functions can be assigned to one Console; a Console Multi-Tasks by dividing its available C+S.

Console Base TL is the ship TL.

B CONSOLES

Console	Tons	Sq	MCr
C Cramped	0.5	1	0.2
S Standard	1	2	0.2
R Roomy	1.5	3	0.2
P Spacious	2	4	0.2

CONSOLE STAFFING

Select a Staffing Level.

C STAFFING LEVELS

Level	Crew Per Console
0 Minimal	1 per 3
1 One Shift	1
2 Two Shifts	2
3 Three Shifts	3

Dangerous= Add one Crew per weapon, defense, sensor, or vehicle, or small craft console.

A **Shift** (a **Watch**) is one-third of a day. Except for Dangerous Voyage, weapons and defenses are unstaffed (they are an additional responsibility).

SHIP'S COMPUTER

Identify Computers by Model Number (equals Cells, or Console Equivalents). Model/0 is little more than a centralized server; Model/5 is five Console-Equivalents. Install Ship's Computers to support the console network.

D STANDARD SHIP COMPUTERS

Model/	Tons	Sq	MCr	TL	Cells
0	0.5	1	0.1	8	0
0 bis	0.5	1	0.5	8	1
1	1	2	1.5	9	1
1 bis	1	2	3.0	9	2
2	2	4	5.0	10	2
2 bis	2	4	7.5	10	3
3	3	6	10.5	11	3
3 bis	3	6	14	11	4
4	4	8	18	12	4
4 bis	4	8	22	12	5
5	5	10	27	13	5
5 bis	5	10	33	13	6
6	6	12	39	14	6
6 bis	6	12	45	14	7
7	7	14	52	15	7
7 bis	7	14	60	15	8
8	8	16	68	16	8
8 bis	8	16	76	16	9
9	9	18	85	17	9

A Ship's Computer may allocate its C+S +K to a number of consoles equal to its Model Number (or Cells).

Tons= Model Number.

Cells= Model Number.

Base TL= Model Number + 8

"Bis" adds +1 Cell.

C+S= TL.

MCr= Model * (Model+.5).

CONSOLES AND CONTROLLERS

Type	C =	Difficulty	Tons	KCr	Comment
CC Control Console	C1 C2 C3 C4 C5	Standard	0.5	200	Includes analog input for piloting.
OC Operating Console	C4 C5	Standard	0.5	100	Standard input by keyboard or similar.
W Workstation	Edu****	Standard	0.5	50	Basic information processing.
M Portable Controller	C4 C5	Difficulty +1D *		100	Laptop equivalent.
H Hand Controller	from User	Difficulty +2D **		50	Tablet equivalent.
CP Control Panel	from User	Difficulty +3D ***			Permanently part of a mechanism.

* M in use occupies one square. ** H is portable by the User. *** CP is on the mechanism.

**** Human devices (based on Education). Others may be based on Instinct or Training.

0.5 ton occupies 1 square (2 cubes) on a deckplan.



Starship Payload 23

B PASSENGER ACCOMMODATIONS

Description	Tons	MCr	Comment
Passenger Commons	1	0.0	For movement and recreation.
Standard Stateroom	2	0.1	1 passenger. No fresher NF
Standard Suite	4	0.2	2 passengers. No fresher
Stateroom Double	2	0.1	2 passengers in bunks. NF.
Stateroom Triple	2	0.1	3 passengers in bunks. NF.
Stateroom Cramped	2	0.1	4 passengers in bunks. NF.
Luxury Stateroom	6	0.4	1 passenger. Includes fresher.
Steerage	0.5	0	2 passengers = Space Bunks
Low Berth	0.5	0.1	One Low Passenger.

C CREW ACCOMMODATIONS

Description	Tons	MCr	Comment
Crew Common Area	1	0.0	For movement and recreation.
Standard Stateroom	2	0.1	1 crew. No fresher.
Standard Crew Suite	4	0.2	2 crew. No fresher
Stateroom Double	2	0.1	2 crew in bunks. NF.
Stateroom Triple	2	0.1	3 crew in bunks. NF.
Stateroom Cramped	2	0.1	4 crew in bunks. NF.
Spacer Niche	1	0.0	1 crew
Spacer Bunks	0.5	0.0	2 crew= Passenger Steerage
Spacer Hotbunks	0.5	0.0	6 crew (2 per shift)
Frozen Watch (also ELB)	2	1.0	contains 10 individuals

D LIFE SUPPORT

Description	Tons	MCr	Comment
Common Fresher	1	1.0	shared by 10 passengers/crew.
Shared Fresher	0.5	0.5	shared by 4 passengers/crew.
Emergency Capsule	1	1.0	LifePod for 10. LS for 15 days.
Emergency Low Berth	1	0.5	contains 4 individuals
Med Console	0.5	0.5	
Medical Low Berth	1	0.5	
Medical Clinic	2	1.0	basic medical supplies
Counsellor	1	0.2	booth with virtual counselling.

E ACCESS

Description	Tons	MCr	Squares	Width	Height
Personnel Air Lock	0.5	0.1	1	1.5	2.5
Cargo Lock	2	0.4	4	3.0	2.5
Large Cargo Lock	9	1.8	18	4.5	2.5

Typical: One personnel Air Lock (no cost / tonnage) per 100 tons of hull

Non-Standard: Create custom Air Locks at MCr0.1 per square.

F CARGO HOLD

Description	Tons	MCr	
Cargo Hold*	1	0.0	Suitable for standard cargo.
Bulk Gas /Liquid Hold	1	0.0	Must be specified.
Bulk Solid Cargo Hold	1	0.0	Granular. Must be specified.
Specialized Hold*	1	0.1	
Sophisticated Hold*	1	1.0	Labs. Non-standard Environ.
Vault*	1	1.0	Required for mail contracts
Ship's Locker*	1	0.0	General support storage.

*Crew Accessible in flight.

A CALCULATE PAYLOAD

Payload is the available space remaining after all other components have been assigned

PAYLOAD

Payload is the total discretionary load the ship can carry in addition to operations requirements. Payload is the available space (tonnage) after all other components have been assigned. In the simplest of ships, this remaining space is undifferentiated Cargo Hold. On other ships, this payload space is fitted out for specific functions, including passenger staterooms, specialized cargo areas, or labs.

- Determine available payload space.
- Allocate passenger accommodations.
- Allocate crew accommodations.
- Allocate required / optional Life Support.
- Assign access and air locks.
- Remaining space as cargo hold.

ACCOMMODATIONS AND LIFE SUPPORT

Assign tonnage for accommodations for passengers and crew. Raw tonnage may be differentiated into staterooms and other locations.

1. High and Middle Passengers. Allocate total tonnage for High Passage (Luxury) and Middle Passage (Standard) accommodations.

2. Low Passage and Steerage. Allocate tonnage for Low Passage (Cold Sleep) and Steerage (Cramped Quarters and Hot Bunks).

LIFE SUPPORT

Commons. Assign Commons as needed.

Fresher. Assign capacity of 1 Fresher per 10 Passenger and Crew.

Medical. Every ship requires a Medical Console. One clinic is required per three Medics, who commonly staff it in watches or shifts.

Access. Assign Air Locks as needed.

FITTED PAYLOAD

Assign fitted spaces for specialized spaces, and sophisticated spaces.

Sophisticated tonnage (routinely used; furnished with custom equipment) such as Labs, Workspaces costs MCr1.0 per ton.

Undifferentiated cargo hold space has no additional cost.

Vault. Secure parcel and precious cargo storage. MCr1 for 1 ton.

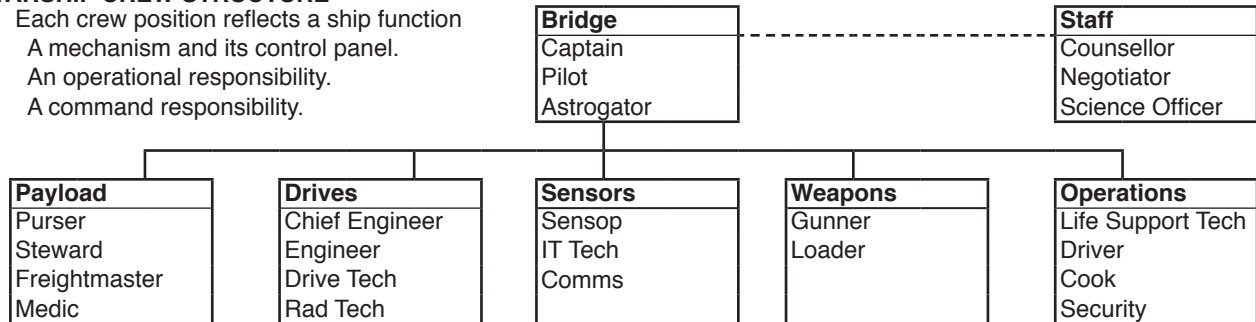


24 Starship Crew Structure

The ship crew is organized in a hierarchy based on abilities and responsibilities. The specific structure depends on the service (such as the Navy) or the mission (profit-making Merchant operations, or on ownership).

STARSHIP CREW STRUCTURE

Each crew position reflects a ship function
A mechanism and its control panel.
An operational responsibility.
A command responsibility.



THE BRIDGE

The Bridge is the command center of the ship, staffed by the Command positions, ship operators, and some staff.

Command Positions

The senior (in rank) crew person is the **Captain** and is in charge of the ship.

The next senior (in rank) is the **Exec** (Executive Officer or First Officer) who is second-in-command.

The Captain and the Exec on small ships may fill other responsibilities (pilot, astrogator).

Ship Operators

The **Pilot** operates the ship's maneuver controls when in normal space. The **Astrogator** computes the ship's course (within a star system and through jump space) and operates the ship's controls when in jump space.

Staff

The Bridge may include (required by individual cultures) staff such as:

A **Counsellor** to evaluate crew and passengers and advise the captain.

A **Political Officer** (or **Cultural Officer**) responsible for loyalty to the government or organization.

A **Specialist** (or **Analyst**) skilled in a specific field: negotiation, diplomacy, liaison, broker, or some other area.

SENSORS

Sensors is the set of scanners, detectors, and comms for the ship. Sensors are co-located with the Bridge.

A **Sensop** operates sensors, communicators, and information systems. **Comms** is a Sensop specializing in communications. An **IT Tech**, if assigned, is usually part of Sensors.

OPERATIONS

Operations is the set of activities for which a ship is designed: research, survey, military operations, or emergency relief, transport, or other activities.

Life Support crew include **Clerk**, **Technician**, **Bookkeeper**, and **Cook**.

A **Driver** or **Vehicle Operator** has responsibility for operation and maintenance of specific vehicles or small craft.

A **Troop** is soldier assigned to internal security and external power projection functions. **Marines** are elite Troops.

Operations also includes various installed facilities: labs, data banks, materials processors

WEAPONS

Weapons covers the combined weapons and defenses of the ship.

A **Gunner** (or, in some cases, a **Sensop**) operates a weapons or defense system. On some weapons, the Gunner is assisted by a **Loader**.

DRIVES

Drives is the propulsion and power section of the ship. An **Engineer** is responsible for operation, maintenance, and repair of drives. Specialists may be assigned: **Rad Tech**, or **Drive Tech**.

The highest ranking Engineer is the **Chief Engineer**.

On smaller ships, an Engineer has responsibilities for all drives; on larger ships, each Engineer specializes on one specific type of drive.

PAYLOAD

Payload is the set of passengers, freight, and specialized installations the ship carries.

A **Steward** is responsible for the welfare of the passengers. The highest ranking Steward is the **Purser**.

A **Freightmaster** is responsible for loading, securing, and offloading freight and cargo.

A **Medic** (or **Doctor**) is responsible for the health and welfare of passengers and crew.

A CREW

Determine total crew members.

B CREW ACCOMMODATIONS

Assign total crew accommodations tonnage.

Equal Opportunity Crews: The appropriate person for specific crew positions is ideally the best skilled person available without regard to species, gender, caste, origin, or other details.

Crew members may (in theory) be sophonts, synthetics, androids, clones, or artificials.



26 Starship Quality

The final ship design is evaluated for its effects on its crew and passengers. Ship designer's decisions influence the opinions of passengers and crew performance.

Calculate Demand, Comfort, and Ease-Of-Use

A. Calculate tonnage for Crew and Passenger Areas:

Passenger Stateroom Tons (23B),

High and Middle Passage

Low Passage and Steerage

Crew Quarters Tons (23C), Ship's Troops (21A) and Common Areas for Crew and Passengers (23D).

B. Determine Passenger Demand (consider Premium Pricing).

C. Determine Crew Comfort (taking into account Tension).

D. Determine Ship Ergonomics (and potential for Mishap).

A CREW / PASSENGER TONNAGE

The tonnage for Passengers (High Passage, Middle Passage, Steerage, and Low Passage) is calculated in 23.

The tonnage for Crew is calculated in 23C.

Ship's Troops is calculated in 21A.

B DEMAND

$$D = (T / P) - 5$$

D= Passage Demand

T= Passenger Quarters Tons. Total tons allocated to passenger accommodations (but not Low and Steerage).

P= Passengers. Total number of available Middle and High passenger accommodations (stateroom beds).

Potential passengers evaluate a ship based on available quarters space and shared recreation space.

Passage Demand determines the allowed Passage Pricing on P Premium Pricing.

P PREMIUM PRICING

Demand	High	Mid	Low
- 5	5,000	3,000	500
- 4	6,000	4,000	600
- 3	7,000	5,000	700
- 2	8,000	6,000	800
- 1	9,000	7,000	900
0	10,000	8,000	1,000
+1	11,000	9,000	1,100
+2	12,000	10,000	1,200
+3	13,000	11,000	1,300
+4	14,000	12,000	1,400
+5	15,000	13,000	1,500

Steerage is double the cost of Low.

Designing for Passenger Demand

Five tons of quarters and commons per passenger produces Demand Level-0.

A ship can be designed for higher or lower premium pricing.

C COMFORT

$$C = (Q / M)$$

C= Crew Comfort (Livability)

Q= Crew Quarters Tons. Total tons allocated to crew accommodations (which includes both common areas and staterooms).

M= Personnel/Crew Members. The total assigned ship crew. If necessary, calculate Ship's Troops and Staff separately.

Crew quarters tonnage significantly influences crew morale, attitudes, and even sanity.

Crew Comfort is a component of the crew's daily Tension roll.

T TENSION

$$T = C + \text{Flux}$$

T= Tension.

C= Comfort

Roll Daily (after 7 days of ship operations). If T is less than Zero, every crew member Check Sanity.

Designing for Crew Comfort

Assign a total tonnage for crew consistent with the desired Livability level. The precise allocation of staterooms and common areas is made when (or if) deck plans are drawn for the ship.

D SHIP ERGONOMICS

$$E = (C / P)$$

E= Ergonomic Quality

C= Total Consoles Tons. Total tonnage for Consoles on the ship.

P= Total Control Panels. Total number of Control Panels on the ship.

Round Up.

Control Ergonomics

Ergonomic environment affects Console users' work quality and their ability to operate without Mishap.

Control Ergonomics is a component of the daily Mishap roll.

M MISHAP

$$M = E + \text{Flux}$$

M= (Potential) Mishap.

E= Ergonomics (Round Up).

Roll Daily. If M is less than Zero (or if unmodified Flux= -5), every Console Check Quality.

Designing for Ergonomics

Allocate redundant Consoles for a Console to Control Panel Ratio of greater than 1. Upgrade Consoles to higher quality. Create roomy control spaces with more tons per Console.



Starship Fillform1

Base TL=

Section	Component	Code	Comment	CP	Sq	Tons	TL	MCr	KCr
02	Mission	Ship Name							
		Mission	E-Mission	D- Qualifier	C- Type	B- Activity	A-Service		
		Mission Modifier	F- Mission Modifier						
04	Hull	A-Hull							
05		C-Configuration							
06		D-Structure							
		E-Fittings							
07	F-Jump Field								
10	Drives	Drive1-Power							
		Drive2-Maneuver							
		Drive3-Interstellar							
		Drive4-							
21	Operations								
22	CP								
23	Payload								
24	Crew	A-Total Crew							
		B-Crew Accommodations and Commons (Tons)							
		C-Passenger Accommodations (Tons)							
		D-Troop Area (Tons)							
Armor Fuel Hardpoints Weapons Defenses Sensors (from FillForm2)									
Totals				CP	Sq	Tons		MCr	KCr

Starship Fillform2



Base TL=

Section	Component	Code	Comment	CP	Sq	Tons	TL	MCr
02	Mission	Ship Name						
		Mission	E-Mission	D-Qualifier	C-Type	B-Activity	A-Service	
		Mission Modifier	F- Mission Modifier					
09	Armor	Armor Layer1						
		Armor Layer2						
		Armor Layer3						
		Armor Coating						
13	Fuel	Fuel1						
		Fuel2						
		Fittings1						
		Fittings2						

Hardpt	Stage	Range R= or S=	Mount	Type	TL	Q	R	E	B	S	CP	Sq	Tons	TL	MCr
Hardpoints	1														
	2														
	3														
	4														
	5														
	6														
	7														
	8														
	9														
	10														
	11														
	12														
Ship Features															
SubTotals											CP	Sq	Tons	TL	MCr



Starship Fillform3

Base TL=

02	Mission	Ship Name					
		Mission	E-Mission	D- Qualifier	C- Type	B- Activity	A-Service
		Mission Modifier	F- Mission Modifier				

24	Crew	A-Total Crew	Bridge Captain Pilot Astrogator		Specialists		Specialists	
		B-Crew Accommodations and Commons (Tons)						
		C-Passenger Accommodations (Tons)						
		D-Troop Area (Tons)						

Operations Security Life Support Drivers 	Drives Power Interstellar InSystem Fuel 	Sensors Comms Computer Sensors 	Weapons Weapons Defenses 	Payload Medic Steward Freight
D Ship's Troops	Passengers	Passengers	Passengers	Passengers

26	Evaluations	D Safe Jump Jump Field Strength S= Drive Efficiency E= $D = \left(\frac{S}{E} \right) - K$ $X = \left(\frac{S}{E} \right) - \left(\frac{D}{K} \right) (+ Flux)$	X Initiation Diameters from Gravity Source D= Jump Drive Engineer S+K K=	D Demand Total Passenger Tons T= Passengers P= $D = \left(\frac{T}{P} \right) - 5$ Passenger Pricing is based on Demand	C Comfort Total Crew Quarters Tons Q= Crew Members M= $C = \left(\frac{Q}{M} \right)$ T Tension $T = + Flux$	E Ergonomics Total Console Tons C= Total Control Panels P= $E = \left(\frac{C}{P} \right)$ M Mishap $M = + Flux$
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